BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2017

PHYSICS

Past Paper 2017 • Paper Code: A

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – C (MULTIPLE CHOICE QUESTIONS)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** A

Note: Q. 16: Choose the correct answer from the given options and write the same in your answer copy.

Q1. Unit of luminous intensity is:

(a) N/C
(b) N/m^2
(c) Jm^2
(d) Candela

Q2. The resultant of two equal and opposite vector is:

(a) Unit vector
(b) Null vector
(c) Position vector
(d) Free vector

Q3. Rate of change of momentum is called:

(a) Torque
(b) Work
(c) Force
(d) Power

Q4. Limiting friction is given by F = μR, where μ is:

(a) Mass per length
(b) Coefficient of viscosity
(b) Coefficient of friction
(d) Weight per length.

Q5. Acceleration of a body moving down a smooth plane inclined at 30° will be:

(a) 980 m/sec^2
(b) 9.8 m/sec^2
(c) 4.9 m/sec^2
(d) 9.4 m/sec^2

Q6. If a body falls freely, the distance covered by it in 2 seconds is:

(a) 9.8 m
(b) 1.96 m
(c) 19.6 m
(d) 9.2 m

Q7. Direction of angular velocity can be determined by:

(a) Law of Sine
(b) Law of Cosine
(c) Right Hand Rule
(d) Left Hand Rule

Q8. At the distance equal to double the earth's radius the value of "g" reduces to:

(a) One half
(b) One third
(c) One fourth
(d) One ninth

Q9. Gravitational P.E possessed by a body lifted through small vertical height "h" is given by:

(a) Mg/h
(b) mh/g
(c) mgh
(d) $1/2\ mv^2$

Q10. Decibel is the unit of:

(a) Intensity of sound
(b) Pitch of sound
(c) Loudness of sound
(d) Intensity level

Q11. Maximum number of beats/seconds that a normal human ear can hear clearly is:

(a) 3 beats /sec
(b) 7 beats/sec
(c) 10 beats/sec
(d) 15 beats/sec

Q12. Beats are produced due to:

(a) Diffraction
(b) Interference
(c) Polarization
(d) Resonance

Q13. "Sone" is the unit of:

(a) Intensity level
(b) Intensity of sound
(c) Pitch of sound
(d) Loudness of sound

Q14. In case of a thin film conditions for path difference for constructive interference is equal to:

(a) $(m + 1/2)\ \lambda$
(b) $m\lambda$
(c) $m\lambda/2$
(d) $(m - 1/2)\ \lambda$

Q15. Polarization of light shows that light waves are:

(a) Electromagnetic
(b) Mechanical
(c) Longitudinal
(d) Transverse

Q16. Wave theory of light cannot explain:

(a) Polarization
(b) Photo electric effect
(c) Interference
(d) Dispersion

Q17. Acceleration of simple pendulum is given by:

(a) $-glx$ (b) $-g/l\ x$

(c) $-l/g\ x$ (d) $-g/2l\ x^2$

SECTION – A (Short Answer Questions)

Time Allowed: 2-40 Hours | **Maximum Marks:** 40

> Note: ATTEMPT ANY EIGHT (08) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q1. Explain stepwise method for addition of Vectors by rectangular components.

Q2. What is fluid friction? State Stoke's Law. Derive an equation for terminal velocity.

Q3. Define centripetal acceleration? Show that $a_c = r\omega^2$.

Q4. Explain the terms " Couple" and " Moment of Couple".

Q5. Explain Artificial gravity. Calculate the frequency of a space craft of tunnel length of 20 m.

Q6. Distinguish between Potential Energy and Kinetic Energy. Derive Work- Energy equation.

Q7. Describe "Beats". Discuss formation of beats.

Q8. Explain how the Law of Vibration of stretched spring is verified by using "Sonometer".

Q9. What is Intensity and loudness of Sound? Describe units for intensity level.

Q10. What is a "Wave front" ? Describe Huygens's Principle for construction of wave fronts.

Q11. Derive magnifying power of magnifying glass.

Q12. How the position, nature and size of an object changes when it is moved from infinity to the optical centre of a Convex lens?

SECTION – B (Long Answer Questions)

Time Allowed: 2-40 Hours | **Maximum Marks:** 28

> Note: ATTEMPT ANY TWO (02) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q13.

(a) [09] State and explain law of conservation of angular momentum. Give the dimensions and units of angular momentum.

(b) [05] A particle of mass 0.4 kg rotates in circular orbit of radius 10 cm at a constant rate of 1.5 rev/sec. Evaluate the angular momentum of that particle with respect to centre of orbit.

Q14.

(a) [09] Define vibratory motion. Discuss the motion of a body under elastic restoring force. Derive an expression for the acceleration of vibrating body under that force.

(b) [05] A body hanging from a spring is set into motion and the period of oscillation is found to be 0.40sec. After the body has come to rest it is removed. How much shorter will be the spring be when it comes to rest?

Q15.

(a) [09] What is Astronomical telescope? Explain its construction and working. Calculate its magnifying power.

(b) [05] An Astronomical telescope of angular magnification 2000 has an objective of 15 m focal length. What is focal length of the eye piece?